Sana Niroomand

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RESEARCH INTERESTS

Computer vision and machine learning for biomedical applications, particularly medical imaging, drug discovery, and generative AI.

EDUCATION

Sharif University of Technology Sept. 2023 – Present
B.S. in Computer Engineering **GPA: 18.94/20.0 (3.79/4.0)**

Selected coursework: Machine Learning (20/20); Intelligent Analysis of Biomedical Images (20/20); Computational Drug Design (19.5/20); Introduction to Bioinformatics (20/20); Artificial Intelligence (18.9/20); Linear Algebra (19.2/20); Deep Generative AI (ongoing).

RESEARCH EXPERIENCE

Research Intern — Object Detection in HDR Images Jul. – Aug. 2026
Middle East Technical University (METU), Ankara, Turkey
Supervisor: Prof. Aydin Alatan [code]

- Studied how images with a wide range of brightness values (high dynamic range, HDR) should be processed for object detection.
- Designed experiments comparing direct HDR inputs with tone-mapped images, which compress brightness values into the range used by standard images.

Related project: Evaluating HDR Image Representations

- Compared **6 image-preprocessing methods** using the same RetinaNet object detector and training settings. Audited **4,080 images and 32,964 object annotations**, then evaluated on a deduplicated split of 2,506 training and 633 test images.
- Reinhard tone mapping increased detection mean average precision (**mAP**) from **24.4 to 31.3 (+6.9 points)** relative to inputs without tone mapping in a single-run comparison.
- Tested whether the finding transferred to a second detector: in a **2-class Faster R-CNN** experiment on 576 test images, Reinhard achieved **44.9 mAP** versus 42.6 for a learned tone-mapping method (RAOD), under matched settings.

SELECTED PROJECTS

Prostate Cancer Prediction from MRI (PI-CAI) [code]

- As the **main contributor**, built a model to identify clinically significant prostate cancer from MRI using exam-level diagnoses as training labels; focused image preprocessing on the prostate region.
- Combined **3 MRI sequences** and image features at **2 spatial scales** using attention-based multiple instance learning (MIL), which aggregates information across slices into one prediction per exam; visualized slice-level attention to inspect model behavior.

DD-PRiSM-plus: Predicting Cancer Drug Responses [code]

- Co-developed a reproducible pipeline to predict how cancer cells respond to individual drugs and drug combinations using NCI60 and NCI-ALMANAC screening data; reproduced published prediction errors within **0.0016 RMSE** across **3 training stages**.
- Combined molecular language-model embeddings (**ChemBERTa**) with chemical structure fingerprints to improve predictions for drugs excluded from training. During single-drug pretraining, increased Pearson correlation (PCC) from **0.7585 to 0.7912** and reduced prediction error (RMSE) by **6.4%**.
- In the same training stage, improved PCC from **0.7673 to 0.7910** when both the drug and cancer cell line were excluded from training; **accelerated training through vectorization** in PyTorch and added numerical-equivalence tests against the reference implementation.

SELECTED PROJECTS (CONTINUED)

Gene-Expression Analysis of Psoriasis and Psoriatic Arthritis [code]

- Reproduced analyses of **2 public RNA-sequencing datasets** to compare gene activity in two related inflammatory diseases. Used R/DESeq2 for differential gene-expression analysis, including **31,362 genes across 48 skin samples** in the psoriasis cohort.
- Matched **16,056 genes** across datasets and observed **Spearman correlation 0.74** between disease-associated expression changes, quantifying similarities between the diseases and examining immune-related patterns.

TEACHING & ACADEMIC SERVICE

Teaching Assistant, Sharif University of Technology 2025 – Present

Probability and Statistics; Intelligent Analysis of Biomedical Images; Introduction to Bioinformatics.

Member, Secretary Team – RSG Iran 2025 – Present

Regional Student Group of Iran, International Society for Computational Biology (ISCB) Student Council.

ACADEMIC AWARDS

- **Top 0.14% nationally** among 147,317 participants in the Mathematics track of the Iranian University Entrance Exam.
- **Gold Medal**, international exhibition in South Korea, for a botanical antimicrobial spray project.
- **2nd Place**, Inventions & Achievements Track, Avicenna Festival.
- **Provincial Distinguished Rank**, Khwarizmi Youth Festival.

LANGUAGES

English: Fluent; IELTS 8 **French:** Conversational **Persian:** Native